

MTWARA, Tanzania, United Republic of

WMO: 639710

Lat: 10.339S

Lon: 40.182E

Elev: 113

StdP: 99.97

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	19.0	14.5	10.4	26.2	15.7	11.3	25.7	11.7	26.3	10.9	25.5	3.1	210	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.9	32.2	25.2	31.9	25.2	31.2	25.1	27.1	30.1	26.7	29.8	26.4	29.4	6.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.9	28.7	26.0	21.6	28.5	25.6	21.2	28.3	86.3	30.3	84.6	29.8	83.2	29.5	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.9	9.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	27.3	27.2	27.1	26.5	25.8	24.9	24.3	24.4	24.8	25.8	27.1	27.6
	DBStd	1.48	1.09	0.95	0.98	0.96	0.76	0.78	0.73	0.90	0.91	1.00	0.84	0.95
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5860	537	481	529	495	446	445	446	443	489	513	545	
	CDD18.3	2818	279	248	271	245	232	196	187	188	193	231	263	287
	CDH23.3	23901	2874	2427	2436	1867	1609	1241	1190	1254	1450	1948	2589	3017
(13) Wind	WSAvg	4.9	5.1	4.7	3.9	4.9	6.0	6.0	6.3	5.2	4.5	4.4	4.0	4.2
	PrecAvg	1167	211	205	230	180	52	13	12	9	12	33	57	160
(15) Precipitation	PrecMax	1685	559	1002	756	389	182	86	62	55	48	164	226	427
	PrecMin	768	22	34	71	47	2	0	0	0	0	0	0	13
	PrecStd	206	129	152	115	78	41	16	15	12	13	37	57	110
	PrecStd	206	129	152	115	78	41	16	15	12	13	37	57	110
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.0	32.4	32.0	31.1	31.0	30.6	31.1	32.0	32.9	33.0	33.0
		MCWB	26.1	26.3	26.3	25.8	24.4	22.7	22.1	21.9	22.7	24.0	24.4	25.6
	2%	DB	31.2	31.2	31.9	31.1	30.2	30.1	29.9	30.2	30.9	31.7	32.1	32.1
		MCWB	26.0	26.1	26.1	25.7	24.2	22.6	21.8	21.9	22.5	23.9	24.6	25.5
	5%	DB	30.9	30.9	31.2	30.7	29.9	29.3	29.1	29.7	30.1	30.8	31.6	31.4
		MCWB	25.9	26.0	26.0	25.5	24.1	22.3	21.7	21.9	22.4	23.7	24.6	25.6
	10%	DB	30.1	30.2	30.6	29.9	29.2	28.9	28.5	28.9	29.2	30.0	31.0	30.9
		MCWB	25.8	25.9	25.9	25.3	23.9	22.2	21.6	21.9	22.5	23.7	24.5	25.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.3	27.6	27.0	26.0	24.7	24.9	24.7	25.0	25.6	26.2	27.2
		MCDB	29.9	30.1	30.8	30.3	29.3	28.6	28.4	29.2	28.8	29.9	30.1	30.3
	2%	WB	26.8	26.7	27.0	26.5	25.4	24.1	23.4	23.7	24.1	25.0	25.8	26.7
		MCDB	29.6	29.7	30.2	29.6	28.8	28.0	27.8	28.0	28.4	29.5	29.7	30.0
	5%	WB	26.4	26.4	26.6	26.1	25.0	23.6	22.7	23.1	23.6	24.5	25.5	26.3
		MCDB	29.2	29.3	29.7	29.2	28.4	27.6	27.2	27.4	28.0	28.9	29.4	29.5
	10%	WB	26.1	26.1	26.2	25.8	24.6	23.1	22.3	22.6	23.1	24.1	25.1	26.1
		MCDB	28.8	29.0	29.3	28.7	27.8	27.1	26.7	26.9	27.6	28.4	29.1	29.2
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.4	7.3	7.0	7.9	8.9	9.6	10.0	10.2	9.4	8.6	6.9
		MCDBR	6.5	7.3	8.2	8.0	8.4	9.4	10.0	10.6	11.0	10.2	9.2	7.8
		MCWBR	2.8	3.1	3.7	3.5	3.5	3.7	3.7	3.9	4.1	3.9	3.2	3.0
	5% WB	MCDBR	6.1	6.6	7.6	7.3	7.7	8.8	9.6	10.0	9.9	9.4	8.4	6.8
		MCWBR	3.0	3.3	3.8	3.6	3.6	3.9	4.2	4.5	4.4	4.0	3.4	3.0
(40) Clear-Sky Solar Irradiance	taub		0.466	0.451	0.411	0.385	0.381	0.370	0.372	0.381	0.395	0.409	0.412	0.425
	taud		2.311	2.372	2.501	2.574	2.556	2.561	2.526	2.484	2.434	2.418	2.432	2.439
	Ebn at Noon		881	890	910	904	879	872	878	896	911	917	923	916
	Edh at Noon		139	130	112	99	96	93	98	107	118	123	122	122
(44) All-Sky Solar Radiation	RadAvg		5.41	5.41	5.29	5.15	5.03	4.83	4.95	5.46	5.97	6.27	6.42	5.93
	RadStd		0.54	0.39	0.38	0.35	0.25	0.16	0.16	0.13	0.16	0.20	0.25	0.42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page